LR-schedule search for CIFAR-10 airbench

Report, 2026-10-16. Code: train_cifar.py (algorithm="auto") and research/. Every run's log is in logs/. "W-n" points to section n of interval_search_writeup.txt, the run-by-run log.

Summary

- **Algorithm:** a multi-pass knot search. Each searched LR is piecewise linear through knots every 20% of training. Pass 1 picks the knots interval by interval, scoring candidates with a lookahead to the end of training. Passes 2-3 redo the sweep using the previous pass's schedule as the lookahead and score candidates on 2 fresh seeds.
- **Batch size 2000:** matches the baseline (93.94) with no transform and under both transforms: 93.88-93.96 when searching one LR (~600 s), 93.81-93.96 when searching all four from scratch (~1800 s).
- **Batch size 500 and 125:** not yet. 0.1-0.25 pp short of the new baselines (94.01 / 93.88). Those runs used the older, weaker baselines; reruns on the new ones are in progress.
- **Main findings:** short lookaheads pick LRs that are too small; the final interval must anneal to 0; single-seed searches suffer a winner's curse; a fixed lookahead shape breaks under transforms, and the previous pass's schedule fixes it.

1. Question

Find one LR-schedule search algorithm that (a) matches the baseline's held-out accuracy, (b) uses as little search time as possible, (c) still works when an unknown transformation is applied on top of the schedule, and (d) still works at batch sizes 500 and 125 (development was at 2000).

Scenario	What is searched	Starting point
A: good baseline	One LR schedule (conv). Also tested: head only; conv + head.	The baseline's own LR; other groups stay on the baseline
B: from scratch	All four trained LRs: conv, head, norm_bias, whiten_bias	Every LR at 1.0; no knowledge of scale or shape

2. Definitions

Term	Meaning
Spec schedule	The LR schedule the search writes (piecewise linear; see section 3).
Transform	A fixed multiplier applied to the spec LR before training uses it, unknown to the search (x = fraction of training). **ramp**: x 4^(2x-1), from x0.25 to x4. **wave**: x 2^sin(2 pi x), x1 -> x2 -> x0.5 -> x1. With a transform the best spec schedule is no longer linear decay (figure 1).
Held-out TTA	Quality of a found schedule: retrained from scratch on seeds 100-107; mean test-set TTA accuracy. Search decisions also use the test set (intentional).
Search seeds	Independent runs of a search; tables give the mean held-out TTA over n of them.
Cost	Wall-clock seconds of the search on one A100, held-out retraining excluded. One full training run at batch size 2000 takes ~10 s.
Lookahead	To score a candidate LR for the current interval, train the interval, then keep training with an assumed future schedule; the TTA at the end is the score.
Do-nothing	The baseline schedule kept unchanged under the transform (no search): what a search must recover.

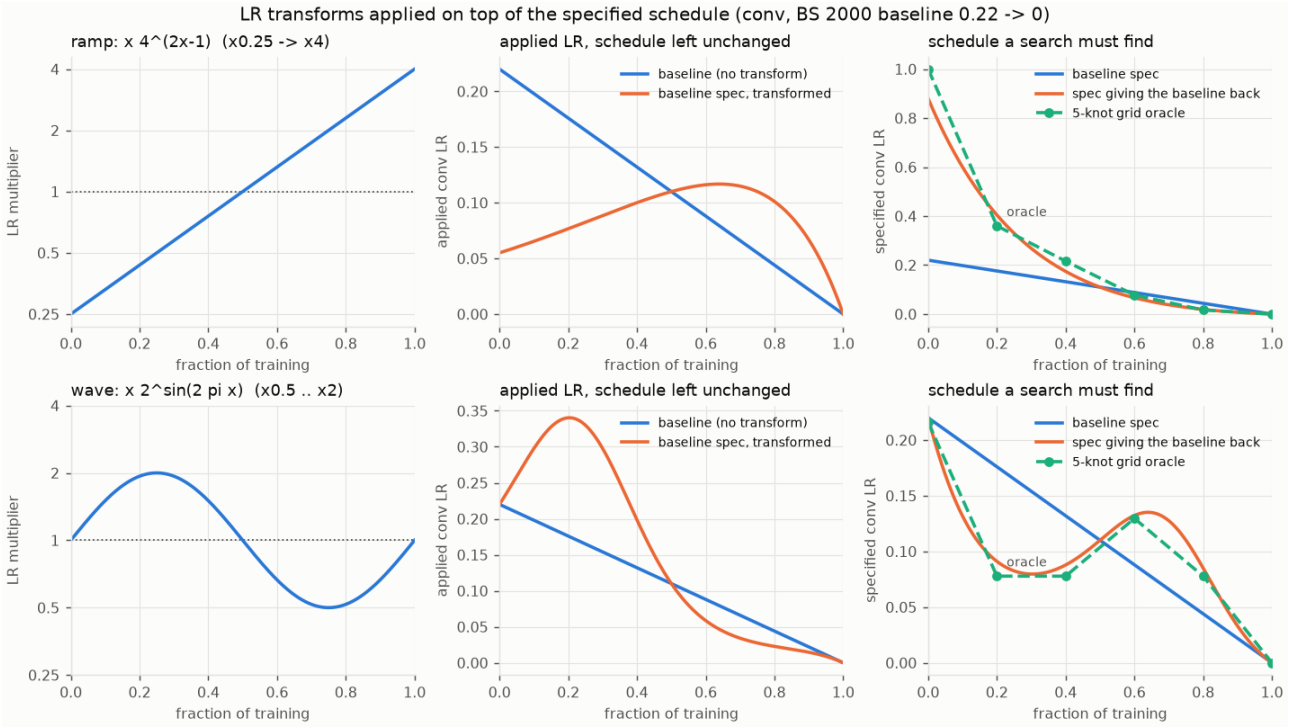


Figure 1. The two transforms (left), the applied conv LR if the baseline's linear decay is kept (middle), and the spec schedule that gives the baseline back with the grid-snapped 5-knot oracle (right). Batch size 2000 conv baseline: 0.22 decaying linearly to 0.

3. Recommended algorithm: multi-pass knot search

Configs: `auto_a_v1x3m2t_*` (scenario A) and `auto_b_v1x2m3fas_*` (scenario B) in train_cifar.py.

Representation

Each searched LR is piecewise linear through 6 knots, one at every 20% of training (5 intervals). Knot values lie on the grid 0.6^k rounded to 2 significant figures.

Pass 1: one forward sweep over the 5 intervals

1. For each searched LR, coordinate-climb the knot at the end of the current interval on the grid, starting from the previous interval's value: try one step up and one step down, keep moving while the score improves.
2. Score a candidate by training the interval from the committed state, then a lookahead to the **end of training** in which the searched LRs decay linearly from the candidate value to 0; the other groups follow their own schedules.
3. Commit the interval with the best knot and continue from the trained state.

Passes 2 and 3: the same sweep, restarted from the initial weights

- Each interval starts its climb at the previous pass's knot.
- The lookahead is the previous pass's schedule for the rest of training, rescaled by candidate / previous knot, so one decision shifts the whole remaining schedule.
- Every candidate is scored as the mean over 2 fresh training seeds.

Additions for scenario B

- First interval: before climbing, scan each LR over 7 values from x10^-3 to x10^3 around its start (with a 20% lookahead) and keep the best.
- Last interval: also try annealing every searched LR linearly to 0; keep it if it scores higher.
- Two passes instead of three; pass 2 scores on 3 seeds and does not rescale the tail.

4. Results

Held-out TTA (%), mean over n search seeds, with search cost in seconds in brackets.

4.1 Scenario A: conv searched, other groups on the baseline

Batch size	Method	No transform	Ramp	Wave
2000	Baseline	93.94		
	Do-nothing		93.42	93.72
	Recommended (3 passes), n=4	**93.96** (588)	**93.93** (611)	**93.88** (572)
	2 passes, n=4	93.98 (359)	93.78 (387)	93.91 (362)
	Pass 1 only, n=2	93.94 (132)	93.57 (150)	93.59 (128)
500	Baseline: old / new	93.83 / 94.01		
	Do-nothing (old baseline)		93.35	93.68
	Recommended, n=2	93.79 (725)	93.75 (774)	93.84 (736)
	Recommended, new baseline	93.92 (751, n=2)	93.95 (743, n=2)	93.96 (730, n=2)
125	Baseline: old / new	93.85 / 93.88		
	Do-nothing (old baseline)		93.31	93.68
	Recommended, n=2 (ramp n=1)	93.64 (1159)	93.88 (1109)	93.72 (1164)
	Recommended, new baseline	running	running	running

Other searched groups (batch size 2000, n=2)	No transform	Ramp	Do-nothing ramp
Head only	93.91	93.90	93.82
Conv + head	93.99	93.73	93.30

4.2 Scenario B: all four LRs from 1.0

Batch size	Baseline	No transform	Ramp	Wave	n
2000	93.94	93.96 (1690)	93.81 (1867)	93.91 (1814)	2
500	94.01	93.89 (2471)	93.90 (2698)	-	1
125	93.88	not run yet			

At batch size 2000, do-nothing under ramp scores 93.25 and one-pass variants 93.05-93.36.

4.3 Verdict

- **Batch size 2000: goal met** in both scenarios and all three conditions (within 0.13 pp, mostly within 0.05 pp), at ~600 s per searched LR (~60 training runs) or ~1800 s from scratch. The earlier 8-seed global search cost 3780 s for four groups and cannot change a schedule's shape.
- **Batch size 500 and 125: not met.** Against the new baselines, scenario A is 0.1-0.25 pp short at 500 and 0.0-0.25 pp short at 125.

4.4 Baseline update

After the campaign, baseline_bs500 and baseline_bs125 were replaced by the best SGD configs of the earlier seed-averaged searches (same optimizers: Muon2 conv, SGD head and biases): batch size 500 94.01 +- 0.12 (seeds 100-115; the previous baseline scored 93.83 on seeds 100-107), batch size 125 93.88 +- 0.11 (previous 93.85, a tie). All batch size 500 / 125 numbers above used the previous baselines, as the target and as the fixed groups of scenario A. Reruns on the new baselines are in progress; do-nothing on them: batch size 500 ramp 93.37, wave 93.77; batch size 125 ramp 93.48, wave 93.69.

5. Findings, each with its data

Bold marks each row's best value. Held-out TTA = mean +- sd over seeds 100-107 unless stated; search rows give the mean over n search seeds (appendix A).

F1. Short lookaheads choose LRs that are too small, whatever the cooldown (W-1, W-4)

Probe at batch size 2000, interval 2 (steps 40-80): train with a constant conv LR during the interval, then either resume the baseline to the end (truth) or run a lookahead and score it. Seed means over 4 seeds. All 72 cooldown designs with a 40-step lookahead (and no lookahead) peak at 0.078 or lower (45 at 0.078, 17 at 0.047, 11 at 0.028); the truth peaks at 0.22; a lookahead to the end of training peaks at 0.13.

Interval-2 conv LR	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Truth (final TTA)	93.63	93.73	93.83	93.94	**93.98**	93.57	93.26
No lookahead	**90.68**	90.65	89.66	87.25	81.99	72.31	55.96
40-step lookahead, 0.22 x LR	91.74	92.36	**92.66**	92.55	91.64	89.37	86.07
40-step lookahead, linear to 0	92.13	92.59	**92.91**	92.59	91.72	90.03	87.97
80-step lookahead, 0.13 x LR	91.90	92.58	93.01	**93.02**	92.73	91.78	90.09
To the end, 0.13 x LR	92.17	92.79	93.21	**93.33**	93.23	92.42	90.80
To the end, linear to 0	93.13	93.52	93.84	**93.88**	93.70	92.82	91.43

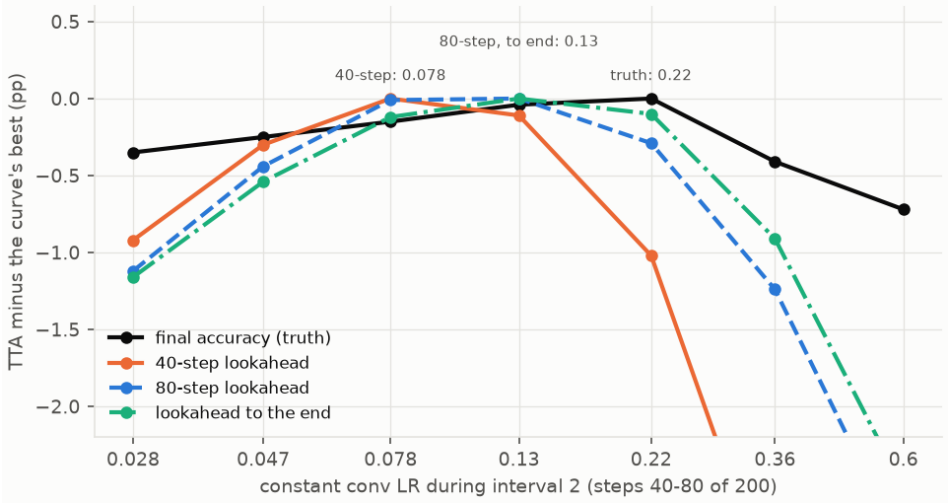


Figure 2. The same data, each curve relative to its own best; labels give each curve's chosen LR.

F2. The non-searched groups' cooldown does not matter (W-1, W-5)

Same probe, 40-step lookahead with the searched LR at 0.22 x candidate, varying what the other groups (head, biases) do during the lookahead. The curves differ by at most 0.1 pp and peak at the same LR. Full searches with the three designs end within 0.05 pp of each other.

Other groups in the lookahead	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Continue their schedule	91.74	92.36	**92.66**	92.55	91.64	89.37	86.07
Decay linearly to 0	91.72	92.33	**92.69**	92.55	91.58	89.41	85.83
Constant 1 x their LR	91.78	92.34	**92.69**	92.56	91.65	89.33	85.90
Constant 0.36 x their LR	91.73	92.34	**92.70**	92.52	91.62	89.29	86.01
Frozen (LR 0)	91.63	92.30	**92.66**	92.46	91.64	89.32	85.47

Full search (20% intervals, searched multiplier)	Held-out TTA	n
Other groups continue	93.78	2
Other groups share the searched multiplier	93.76	2
Other groups get their own searched multiplier	93.73	2

F3. The final interval must anneal to 0; constant intervals are otherwise fine (W-6)

Fixed schedules on held-out seeds (no search). The conv schedule of baseline_bs2000 (0.22 decaying linearly to 0) is replaced by n constant steps at the mean of each step, optionally snapped to the 0.6^k grid; the last two rows keep 5 constant steps but let the last one decay to 0.

Conv schedule (batch size 2000)	Held-out TTA
Baseline: linear decay 0.22 -> 0	93.92 +- 0.13
5 constant steps	93.64 +- 0.07
5 constant steps, grid-snapped	93.52 +- 0.11
10 constant steps	93.92 +- 0.19
20 constant steps	93.95 +- 0.09
5 steps, last decays linearly (same mean)	93.90 +- 0.10
5 steps, last is the baseline's own (0.044 -> 0)	93.91 +- 0.12
All four groups as 5 constant steps	93.68 +- 0.19

Grid-snapped 5-knot piecewise-linear baseline	Held-out TTA	That batch size's baseline
Batch size 2000	93.97 +- 0.11	93.92 +- 0.13
Batch size 500	93.84 +- 0.11	93.83 +- 0.12 (old)
Batch size 125	93.83 +- 0.14	93.85 (old; seeds 100-107)

F4. A fixed lookahead shape breaks under transforms (W-7, W-8, W-11)

Pass 1 alone (linear-to-0 lookahead) recovers little under the transforms, and under wave ends below doing nothing. Its committed knots under ramp stay far below the oracle. Alternative lookahead shapes trade one condition for another.

Batch size 2000, scenario A	None	Ramp	Wave
Baseline / do-nothing	93.94	93.42 +- 0.15	93.72 +- 0.13
Pass 1 only (linear lookahead)	93.94	93.57	93.59
Searched lookahead level (plan)	93.82	93.47	93.59
Geometric lookahead, searched ratio	93.74	93.75	93.85
Max of linear and geometric	93.88	93.71	93.76

Conv knots under ramp (start, 20%, ..., end)	Knots
Oracle (grid-snapped baseline / ramp)	1, 0.36, 0.22, 0.078, 0.017, 0
Pass 1 only, search seed 0	0.13, 0.13, 0.13, 0.028, 0.017, 0.0022

F5. The previous pass's schedule is a self-correcting lookahead (W-12, W-19)

Under ramp, each extra pass raises the early knots toward the oracle and the held-out TTA toward the baseline.

Batch size 2000, scenario A	None	Ramp	Wave	n
1 pass	93.94	93.57	93.59	2
2 passes (pass 2 on the same seed)	93.92	93.94	93.50	2
2 passes, 2 fresh seeds, tail rescaled	93.98	93.78	93.91	4
3 passes, 2 fresh seeds, tail rescaled	93.96	93.93	93.88	4

Conv knots under ramp	Knots
Oracle	1, 0.36, 0.22, 0.078, 0.017, 0
1 pass (seed 0)	0.13, 0.13, 0.13, 0.028, 0.017, 0.0022
2 passes (seed 0)	0.6, 0.6, 0.22, 0.047, 0.017, 0.0005
3 passes (seed 1)	0.6, 0.6, 0.2127, 0.0467, 0.01, 0.0008

F6. Single-seed searches suffer a winner's curse (W-12, W-13, W-16)

In a two-pass search whose second pass reuses the first pass's seed, re-scoring the incumbent replays pass 1 exactly, so pass 2 inherits a lucky score and never moves (wave, search seed 1: the same 94.09 in every pass-2 interval; held out, this schedule scores about 93.4). Scoring pass 2 on fresh seeds removes the lock-in.

Interval	Pass 1: score of the chosen knot	Pass 2: score of the chosen knot
1	93.82 (knot 0.13)	94.09 (knot 0.13)
2	93.98 (knot 0.13)	94.09 (knot 0.13)
3	93.95 (knot 0.13)	94.09 (knot 0.13)
4	93.82 (knot 0.13)	94.09 (knot 0.13)
5	94.09 (knot 0.01)	94.09 (knot 0.01)

Wave, batch size 2000	Own score	Held-out	Optimism
2 passes, pass 2 on the same seed	93.92	93.50	+0.42
2 passes, pass 2 on 1 fresh seed	93.88	93.70	+0.18
2 passes, pass 2 on 3 fresh seeds	93.93	93.83	+0.10
... plus tail rescaling	94.00	93.89	+0.11

F7. A few training steps reveal the conv LR scale, but not the bias LRs (W-2, W-10, W-16)

Each group's constant LR swept over a x sqrt(10) grid (others at their baseline LR), loss on 2000 test images after 16 training steps, mean of 2 seeds. Conv has a sharp minimum at 0.1 for every batch size; the bias curves are nearly flat. Over a full run, though, norm_bias matters and whiten_bias does not.

Group	Batch size	Best LR after 16 steps	Loss rise one x3.2 step below / above	Full-run LR (old baseline)
conv	2000	0.1	0.162 / 0.166	0.22
conv	500	0.1	0.180 / 0.053	0.13
conv	125	0.1	0.106 / 0.021	0.047
head	2000	1000	0.062 / 0.018	1300
head	500	1000	0.055 / 0.030	280
head	125	3200	0.004 / 0.260	170
norm_bias	2000	320	0.018 / 0.046	99
norm_bias	500	1000	0.018 / 0.032	21
norm_bias	125	1000	0.013 / 0.043	4.6
whiten_bias	2000	320	0.008 / 0.041	36
whiten_bias	500	1000	0.006 / 0.006	36
whiten_bias	125	320	0.005 / 0.006	7.7

baseline_bs2000 with ...	Held-out TTA
nothing changed	93.92 +- 0.13
whiten_bias LR 36 -> 1	93.92 +- 0.15
norm_bias LR 99 -> 1	93.66 +- 0.14
both	93.61 +- 0.14

F8. The representation is not the limit under transforms (W-11)

Conv knots set to baseline / transform at each 20% boundary (exact, or snapped to the grid), linear in between, trained under the transform.

Batch size 2000	Exact knots	Grid-snapped knots	Do-nothing	Baseline
Ramp	93.86 +- 0.10	93.91 +- 0.26	93.42 +- 0.15	93.94
Wave	93.97 +- 0.17	93.97 +- 0.20	93.72 +- 0.13	93.94

6. What did not work

Idea	Result	Log
Searched cooldown multiplier, 20% lookahead	best 93.78 at batch size 2000	W-5
Prefer a larger LR within x pp of the best	93.54-93.65	W-5
Search the lookahead's level / a geometric shape	93.82 / 93.74 without a transform	W-8
Max of linear and geometric lookahead	93.88 / 93.71 / 93.76	W-11
Full-run knot polish on a single seed	fixes ramp, not wave	W-9
Sloped first segment (first_knot)	wave 93.45-93.61	W-12
Cheap pass 1 (20% lookahead), then pass 2	93.64	W-17
Freeze LRs that look flat in the first scan	scenario B ramp 93.13	W-16

7. Open problems

- Batch sizes 500 and 125 fall short of the new baselines. The gap is in the search, not the representation (F3). At batch size 125 the search's own score exceeds its held-out score by about 0.2 pp, a sign of remaining winner's curse.
- The spread between search seeds is 0.1-0.4 pp, about the size of the differences between the best variants; 2-4 search seeds rank them only coarsely.
- Scenario B costs about 3x scenario A, and cost roughly doubles from batch size 2000 to 125.
- Only multiplicative transforms were tested.

8. Reproduce

Task	Command
Run a search	CIFAR_EXPERIMENTS=auto_a_v1x3m2t_bs2000_ramp CIFAR_SEEDS=0 /venv/main/bin/python train_cifar.py (the log ends with auto_result and heldout_result lines)
Summarize runs	python research/summarize_auto.py [--since YYYYMMDD_HHMMSS] [name ...]
Comparison table	python research/final_table.py
Probes	research/cooldown_fidelity.py (F1, F2), research/overfit_screen.py (F7)
Figures	python research/plot_transforms.py; python research/plot_lookahead_bias.py
This report	python research/make_report.py

Appendix A. Every search run (from logs/)

One row per search config: n search seeds, mean search cost, mean evaluations, the search's own final TTA, the held-out TTA (seeds 100-107) and its range over search seeds. A.1: runs with the previous batch size 500 / 125 baselines (before 2026-10-16 18:30 UTC). A.2: runs after the baseline update.

A.1 Before the baseline update

Search config	n	Cost (s)	Evals	Own	Held-out	Range
auto_a2_v1x2m3t_bs2000	2	938	78	94.03	93.99	0.14
auto_a2_v1x2m3t_bs2000_ramp	2	903	77	93.90	93.73	0.02
auto_a_lin_rest_lin_bs2000	2	132	22	93.99	93.94	0.02
auto_a_lin_ssearch_bs2000	2	140	34	93.75	93.85	0.07
auto_a_nocool_bs2000	2	62	22	92.20	92.12	0.46
auto_a_rest_lin_bs2000	2	130	22	93.77	93.82	0.00
auto_a_rest_s013_bs2000	2	129	21	93.90	93.80	0.00
auto_a_rest_s013_fa_bs2000	2	126	20	93.93	93.88	0.06
auto_a_rest_s022_bs2000	2	135	22	93.79	93.72	0.26
auto_a_rest_ssearch_bs2000	2	223	34	93.84	93.70	0.02
auto_a_s036_cont_bs2000	2	90	22	93.51	93.38	0.33
auto_a_s036_cont_lw1_bs2000	2	96	23	93.66	93.54	0.01
auto_a_s036_cont_lw2_bs2000	2	101	24	93.66	93.65	0.22
auto_a_s036_shared_bs2000	2	91	22	93.62	93.50	0.18
auto_a_s20x2m3t_bs2000	2	448	40	93.82	93.64	0.52
auto_a_s20x2m3t_bs2000_ramp	2	451	44	93.85	93.76	0.20
auto_a_ssearch_cont_bs2000	2	127	30	93.71	93.78	0.02
auto_a_ssearch_fa_bs2000	2	122	28	93.79	93.88	0.09
auto_a_ssearch_fas_bs2000	2	137	35	93.80	93.86	0.13
auto_a_ssearch_osearch_bs2000	2	163	39	93.69	93.73	0.20
auto_a_ssearch_shared_bs2000	2	130	31	93.68	93.76	0.38
auto_a_v1_bs2000_ramp	2	150	26	93.62	93.57	0.05
auto_a_v1_bs2000_wave	2	128	22	93.91	93.59	0.03
auto_a_v1k_bs2000	2	152	24	94.03	93.87	0.08
auto_a_v1k_bs2000_ramp	2	179	29	93.82	93.73	0.09
auto_a_v1k_bs2000_wave	2	182	28	93.84	93.61	0.31
auto_a_v1p2_bs2000	2	481	56	94.13	94.00	0.10
auto_a_v1p2_bs2000_ramp	2	453	54	93.87	93.75	0.11
auto_a_v1p2_bs2000_wave	2	440	53	94.19	93.75	0.35
auto_a_v1p_bs2000	2	267	35	94.08	93.93	0.05
auto_a_v1p_bs2000_ramp	2	369	48	94.13	93.92	0.11
auto_a_v1p_bs2000_wave	2	259	35	93.91	93.59	0.03
auto_a_v1p_bs500	2	335	36	94.05	93.81	0.07
auto_a_v1p_bs500_ramp	2	405	44	93.98	93.80	0.08
auto_a_v1pk_bs2000	1	282	37	94.04	93.83	0.00
auto_a_v1pk_bs2000_ramp	2	311	42	93.82	93.73	0.09
auto_a_v1pk_bs2000_wave	2	342	44	93.90	93.58	0.26
auto_a_v1pr_bs2000	2	272	36	94.11	93.94	0.02
auto_a_v1pr_bs2000_ramp	2	459	57	93.91	93.81	0.01
auto_a_v1pr_bs2000_wave	2	268	36	93.77	93.59	0.03
auto_a_v1x2_bs2000	2	233	36	93.99	93.92	0.07

Search config	n	Cost (s)	Evals	Own	Held-out	Range
auto_a_v1x2_bs2000_ramp	2	272	44	94.19	93.94	0.07
auto_a_v1x2_bs2000_wave	2	229	37	93.92	93.50	0.15
auto_a_v1x2m2t_bs125	2	712	39	93.79	93.63	0.09
auto_a_v1x2m2t_bs125_ramp	2	722	42	93.72	93.65	0.00
auto_a_v1x2m2t_bs125_wave	2	732	42	93.86	93.79	0.02
auto_a_v1x2m2t_bs2000	4	359	39	94.08	93.98	0.19
auto_a_v1x2m2t_bs2000_ramp	4	387	44	93.83	93.78	0.43
auto_a_v1x2m2t_bs2000_wave	4	362	40	94.03	93.91	0.03
auto_a_v1x2m2t_bs500	2	446	39	93.97	93.83	0.10
auto_a_v1x2m2t_bs500_ramp	2	500	46	93.75	93.67	0.24
auto_a_v1x2m2t_bs500_wave	2	456	40	94.00	93.91	0.12
auto_a_v1x2m3_bs125	2	873	38	93.87	93.81	0.03
auto_a_v1x2m3_bs125_ramp	2	886	40	93.80	93.75	0.26
auto_a_v1x2m3_bs125_wave	2	905	40	93.72	93.50	0.30
auto_a_v1x2m3_bs2000	2	444	37	94.02	93.95	0.11
auto_a_v1x2m3_bs2000_ramp	2	495	44	93.93	93.86	0.21
auto_a_v1x2m3_bs2000_wave	2	494	40	93.93	93.83	0.14
auto_a_v1x2m3_bs500	2	551	38	93.97	93.91	0.02
auto_a_v1x2m3_bs500_ramp	2	658	44	94.02	93.85	0.00
auto_a_v1x2m3_bs500_wave	2	624	41	93.91	93.85	0.27
auto_a_v1x2m3t_bs125	2	970	40	93.78	93.64	0.14
auto_a_v1x2m3t_bs125_ramp	2	949	42	93.73	93.66	0.03
auto_a_v1x2m3t_bs125_wave	2	912	40	93.86	93.75	0.07
auto_a_v1x2m3t_bs2000	4	467	39	94.01	93.96	0.20
auto_a_v1x2m3t_bs2000_ramp	4	513	44	93.81	93.78	0.45
auto_a_v1x2m3t_bs2000_wave	4	480	40	94.00	93.89	0.14
auto_a_v1x2m3t_bs500	2	599	40	93.97	93.89	0.03
auto_a_v1x2m3t_bs500_ramp	2	649	45	93.82	93.79	0.01
auto_a_v1x2m3t_bs500_wave	2	609	40	93.99	93.92	0.12
auto_a_v1x2r_bs2000	2	237	38	94.18	93.98	0.05
auto_a_v1x2r_bs2000_ramp	2	257	42	94.08	93.80	0.03
auto_a_v1x2r_bs2000_wave	2	241	39	93.88	93.70	0.19
auto_a_v1x3m2t_bs125	2	1159	57	93.83	93.64	0.10
auto_a_v1x3m2t_bs125_ramp	1	1109	58	93.91	93.88	0.00
auto_a_v1x3m2t_bs125_wave	2	1164	60	93.97	93.72	0.12
auto_a_v1x3m2t_bs2000	4	588	57	93.96	93.96	0.19
auto_a_v1x3m2t_bs2000_ramp	4	611	62	94.14	93.93	0.13
auto_a_v1x3m2t_bs2000_wave	4	572	56	94.00	93.88	0.20
auto_a_v1x3m2t_bs500	2	725	57	93.91	93.79	0.23
auto_a_v1x3m2t_bs500_ramp	2	774	62	93.93	93.75	0.19
auto_a_v1x3m2t_bs500_wave	2	736	58	93.97	93.84	0.19
auto_a_v2plan_bs2000	2	214	34	94.09	93.82	0.23
auto_a_v2plan_bs2000_ramp	2	254	38	93.51	93.47	0.38
auto_a_v2plan_bs2000_wave	1	211	36	93.84	93.59	0.00
auto_a_v3_bs2000	2	215	35	93.84	93.74	0.10
auto_a_v3_bs2000_ramp	2	242	39	93.95	93.75	0.07

Search config	n	Cost (s)	Evals	Own	Held-out	Range
auto_a_v3_bs2000_wave	2	184	29	94.06	93.85	0.20
auto_a_v3alt_bs2000	2	250	40	93.84	93.74	0.10
auto_a_v3alt_bs2000_ramp	2	275	44	93.95	93.75	0.07
auto_a_v3alt_bs2000_wave	2	218	32	94.06	93.85	0.20
auto_a_v3p_bs2000	2	351	48	93.92	93.76	0.07
auto_a_v3p_bs2000_ramp	2	374	52	93.95	93.75	0.07
auto_a_v3p_bs2000_wave	2	315	42	94.06	93.85	0.20
auto_a_v4_bs2000	2	330	31	93.94	93.88	0.10
auto_a_v4_bs2000_ramp	2	395	37	93.86	93.71	0.14
auto_a_v4_bs2000_wave	2	307	30	93.95	93.76	0.02
auto_ahead_v1x2m3t_bs2000	2	446	33	94.11	93.91	0.00
auto_ahead_v1x2m3t_bs2000_ramp	2	457	36	93.98	93.90	0.13
auto_b_v1_bs2000	2	588	84	93.99	93.72	0.29
auto_b_v1_bs2000_ramp	2	648	92	93.37	93.16	0.15
auto_b_v1_bs2000_wave	2	701	104	94.08	93.70	0.12
auto_b_v1fas_bs2000	2	635	118	94.25	93.92	0.10
auto_b_v1fas_bs2000_ramp	2	642	129	93.36	93.20	0.24
auto_b_v1fast_bs2000	2	618	110	94.09	93.90	0.02
auto_b_v1fast_bs2000_ramp	2	623	120	93.31	93.05	0.24
auto_b_v1ps_bs2000	2	1158	145	94.10	93.86	0.01
auto_b_v1ps_bs2000_ramp	2	1313	170	93.43	93.12	0.49
auto_b_v1psfast_bs2000	2	1066	154	94.16	93.95	0.07
auto_b_v1psfast_bs2000_ramp	2	1058	164	93.58	93.36	0.19
auto_b_v1x2m2fasff_bs2000	1	1153	146	93.93	93.66	0.00
auto_b_v1x2m2fasff_bs2000_ramp	2	915	111	93.35	93.13	0.78
auto_b_v1x2m3fas_bs2000	2	1690	168	94.13	93.96	0.02
auto_b_v1x2m3fas_bs2000_ramp	2	1867	188	94.01	93.81	0.08
auto_b_v1x2m3fas_bs2000_wave	2	1814	176	94.07	93.91	0.00
auto_b_v1x2m3fas_bs500	1	2471	191	94.20	93.89	0.00
auto_b_v1x2m3fas_bs500_ramp	1	2698	200	94.13	93.90	0.00
auto_b_v1x2m3tfas_bs2000_ramp	1	1845	184	93.73	93.52	0.00
auto_b_v1x2m3tfas_bs2000_wave	1	1665	187	94.11	93.92	0.00

A.2 After the baseline update

Search config	n	Cost (s)	Evals	Own	Held-out	Range
auto_a_v1x3m2t_bs500	2	751	58	93.99	93.92	0.09
auto_a_v1x3m2t_bs500_ramp	2	743	61	94.11	93.95	0.02
auto_a_v1x3m2t_bs500_wave	2	730	56	94.03	93.96	0.10

Appendix B. Every fixed-schedule run on held-out seeds (from logs/)

"old" / "new" marks batch size 500 / 125 runs made with the previous / updated baselines.

Fixed schedule (no search)	Baselines	Seeds	Held-out TTA
baseline_bs125	old	100-103	93.80 +- 0.10
baseline_bs125	old	104-107	93.90 +- 0.07
baseline_bs2000	-	100-107	93.92 +- 0.13
baseline_bs2000	-	108-123	93.96 +- 0.12

Fixed schedule (no search)	Baselines	Seeds	Held-out TTA
baseline_bs2000_nb1	-	100-107	93.66 +- 0.14
baseline_bs2000_nb1wb1	-	100-107	93.61 +- 0.14
baseline_bs2000_wb1	-	100-107	93.92 +- 0.15
baseline_bs500	old	100-107	93.83 +- 0.12
oracle_knots_ramp_bs2000	-	100-107	93.86 +- 0.10
oracle_knots_ramp_grid_bs2000	-	100-107	93.91 +- 0.26
oracle_knots_wave_bs2000	-	100-107	93.97 +- 0.17
oracle_knots_wave_grid_bs2000	-	100-107	93.97 +- 0.20
schedule_global_interval_conv_lr_bs2000	-	100-107	93.72 +- 0.14
schedule_global_interval_lrs_bs2000	-	100-107	93.85 +- 0.12
schedule_interval_conv_lr_bs2000	-	100-107	93.77 +- 0.14
schedule_interval_lrs_bs2000	-	100-107	93.92 +- 0.10
seeded_best_bs125_sgd	-	100-115	93.88 +- 0.16
seeded_best_bs500_sgd	-	100-115	94.01 +- 0.12
staircase_a_knots_grid_bs125	old	100-107	93.83 +- 0.14
staircase_a_knots_grid_bs2000	-	100-107	93.97 +- 0.11
staircase_a_knots_grid_bs500	old	100-107	93.84 +- 0.11
staircase_a_lastexact_bs2000	-	100-107	93.91 +- 0.12
staircase_a_lastlin_bs2000	-	100-107	93.90 +- 0.10
staircase_a_n10_bs2000	-	100-107	93.92 +- 0.19
staircase_a_n20_bs2000	-	100-107	93.95 +- 0.09
staircase_a_n5_bs2000	-	100-107	93.64 +- 0.07
staircase_a_n5_grid_bs2000	-	100-107	93.52 +- 0.11
staircase_b_n5_bs2000	-	100-107	93.68 +- 0.19
staircase_b_n5_grid_bs2000	-	100-107	93.64 +- 0.15
transformed_a2_ramp_bs2000	-	100-107	93.30 +- 0.10
transformed_a_ramp_bs125	new	100-107	93.48 +- 0.12
transformed_a_ramp_bs125	old	100-107	93.31 +- 0.09
transformed_a_ramp_bs2000	-	100-107	93.42 +- 0.15
transformed_a_ramp_bs500	new	100-107	93.37 +- 0.14
transformed_a_ramp_bs500	old	100-107	93.35 +- 0.17
transformed_a_wave_bs125	new	100-107	93.69 +- 0.07
transformed_a_wave_bs125	old	100-107	93.68 +- 0.16
transformed_a_wave_bs2000	-	100-107	93.72 +- 0.13
transformed_a_wave_bs500	new	100-107	93.77 +- 0.18
transformed_a_wave_bs500	old	100-107	93.68 +- 0.15
transformed_b_ramp_bs2000	-	100-107	93.25 +- 0.11
transformed_b_wave_bs2000	-	100-107	93.68 +- 0.11
transformed_head_ramp_bs2000	-	100-107	93.82 +- 0.16